solid confirmation of the standard cosmological model, based on general relativity with the cosmological constant. The detection of gravitational waves is a spectacular confirmation of general relativity, a theory a hundred years old. The three results, obtained with strenuous technological efforts and extensive collaborations between hundreds of scientists, do nothing other than reinforce the understanding that we already had of the structure of the universe. No real surprises.

But such an absence of surprises was in a sense itself surprising, because many expected to be surprised, that is, to see ‘new physics’, not yet described by established theories. They expected supersymmetry at CERN, not the Higgs boson. And many expected that Planck would measure discrepancies from the standard cosmological model, discrepancies that would support alternative cosmological theories to general relativity.

But no. What nature is confirming is simple: general relativity, quantum mechanics and, within quantum mechanics, the standard model.

Many theoretical physicists are today looking for new theories by picking arbitrary hypotheses. ‘Let us imagine that …’ I don’t think that this way of doing science has ever produced good results. Our fantasy is too limited to ‘imagine’ how the world may be made, unless we search for inspiration in the traces we have at our disposal. The traces that we have – our clues – are either the theories which have been successful, or new experimental data, nothing else. It is in this data and in these theories that we must try to uncover what we have been unable yet to imagine. This is how Copernicus, Newton, Maxwell and Einstein proceeded. They never tried to ‘guess’ a new theory – unlike, in my opinion, the way in which too many theoretical physicists are trying to do today.

The three recent experimental results I mentioned speak with the voice of Nature itself: ‘Stop dreaming of new fields and strange particles; supplementary dimensions, other symmetries, parallel universes, strings, and whatever else. The pieces of the puzzle are simpler: general relativity, quantum mechanics and the standard model. The next step forward may be “only” a question of combining them in the correct manner.’ It’s reassuring advice for the loop quantum gravity community, because these are the hypotheses of the theory: general relativity, quantum mechanics and compatibility with the standard model, nothing else. The radical conceptual consequences – the quanta of space, the disappearance of time – are not bold hypotheses: they are the rational consequences that follow from taking the basic insights of our best theories seriously.

Once again, these are not definitive proofs. Supersymmetric particles might finally exist, perhaps, at a scale still not reached, and could exist even if loop theory is correct. Supersymmetry failed to show up where expected, and